

EXPERIMENT 8

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Subject Name: ADBMS

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Experiment 8 -- To study and implement Views

Aim : To study and learn the about the Views, grant or revoke submission.

Question :8 Part A (i): Complex View

SQL Query:

```
CREATE TABLE customer_master
```

```
(  
    customer_id      VARCHAR(5)  
    PRIMARY KEY,      full_name  
    VARCHAR(50) NOT NULL,    city  
    VARCHAR(30)  
);
```

```
CREATE TABLE product_catalog
```

```
(  
    product_id      VARCHAR(5)  
    PRIMARY KEY,      product_name  
    VARCHAR(50) NOT NULL,
```

EXPERIMENT 8

unit_price NUMERIC(10,2) NOT
NULL

);

CREATE TABLE sales_orders

(

order_id SERIAL PRIMARY KEY,

product_id VARCHAR(5)

REFERENCES product_catalog(product_id),

quantity INT NOT NULL,

customer_id VARCHAR(5)

REFERENCES customer_master(customer_id),

discount_percent NUMERIC(5,2),

order_date DATE NOT NULL

);

INSERT INTO customer_master

(customer_id, full_name, city)

VALUES

EXPERIMENT 8

('C1', 'Amit Sharma', 'Delhi'),
('C2', 'Priya Verma', 'Mumbai'),
('C3', 'Ravi Kumar', 'Bangalore'),
('C4', 'Neha Singh', 'Kolkata'),
('C5', 'Arjun Mehta', 'Hyderabad');

```
INSERT INTO product_catalog
(product_id, product_name, unit_price)
VALUES
('P1', 'Smartphone X100', 25000),
('P2', 'Laptop Pro 15', 65000),
('P3', 'Wireless Earbuds', 5000),

('P4', 'Smartwatch Fit', 30000),
('P5', 'Gaming Console', 45000);
```

```
INSERT INTO sales_orders
(product_id, quantity, customer_id, discount_percent, order_date)
VALUES
('P1', 2, 'C1', 5, '2025-09-01'),
('P2', 1, 'C2', 10, '2025-09-02'),
('P3', 3, 'C3', 0, '2025-09-03'),
('P4', 1, 'C4', 8, '2025-09-04'),
```

EXPERIMENT 8

```
('P5', 1, 'C5', 12, '2025-09-05'),  
( 'P1', 1, 'C2', 5, '2025-09-06'),  
( 'P3', 2, 'C1', 0, '2025-09-07');
```

```
CREATE OR REPLACE VIEW vw_sales_report as  
select  t.order_id, cc.full_name,t.quantity, t.final_am as final_amount  
FROM (  
    select so.discount_percent, so.product_id, so.customer_id,so.order_id, so.quantity, (  
        (pc.unit_price * so.quantity * so.discount_percent )/100  
        ) as final_am  
    FROM sales_orders as so  
    JOIN product_catalog as pc  
    ON so.product_id = pc.product_id  
) t  
JOIN customer_master as cc  
ON cc.customer_id =t.customer_id  
  
SELECT * from vw_sales_report  
  
CREATE USER Exp_8_user with PASSWORD '1234';  
GRANT SELECT ON vw_sales_report TO Exp_8_user;
```

EXPERIMENT 8

Question: Part A (ii) Materialized view for Category_report_generation SQL
Query:

```
DROP TABLE IF EXISTS category_master CASCADE;  
DROP TABLE IF EXISTS product_master CASCADE;  
DROP TABLE IF EXISTS customer_master CASCADE;  
DROP TABLE IF EXISTS order_details CASCADE;
```

```
CREATE TABLE category_master (  
category_id          VARCHAR(5)  
PRIMARY KEY,        category_name  
VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE product_master (  
product_id VARCHAR(5)  
PRIMARY KEY, product_name  
VARCHAR(50) NOT NULL,  
category_id VARCHAR(5)  
REFERENCES category_master(category_id)  
);
```

EXPERIMENT 8

```
CREATE TABLE customer_master (  
  
    customer_id VARCHAR(5) PRIMARY KEY,  
    customer_name VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE order_details  
(    order_id VARCHAR(5)  
NOT NULL,    customer_id  
VARCHAR(5)  
REFERENCES  
customer_master(customer_id),    product_id  
VARCHAR(5)  
REFERENCES  
product_master(product_id),  
quantity INT NOT NULL,  
unit_price NUMERIC(10,2) NOT  
NULL  
);
```

```
INSERT INTO category_master  
(category_id, category_name)
```

EXPERIMENT 8

VALUES

('C1', 'Electronics'),

('C2', 'Clothing'),

('C3', 'Books');

INSERT INTO product_master

(product_id, product_name, category_id)

VALUES

('P101', 'Laptop', 'C1'),

('P102', 'Headphones', 'C1'),

('P201', 'Jacket', 'C2'),

('P202', 'Shoes', 'C2'),

('P301', 'SQL Book', 'C3');

INSERT INTO customer_master

(customer_id, customer_name)

VALUES

('U1', 'Aman'),

('U2', 'Priya'),

('U3', 'Rahul'),

('U4', 'Simran');

INSERT INTO order_details

EXPERIMENT 8

(order_id, customer_id, product_id, quantity, unit_price)

VALUES

('O101', 'U1', 'P101', 1, 5000),
('O102', 'U2', 'P102', 2, 1000),
('O103', 'U1', 'P201', 1, 3000),
('O104', 'U3', 'P202', 2, 2000),
('O105', 'U4', 'P301', 3, 500),
('O106', 'U2', 'P101', 1, 5000),
('O107', 'U3', 'P201', 1, 3000),
('O108', 'U4', 'P102', 1, 1000);

DROP MATERIALIZED VIEW mw_category_report

EXPLAIN ANALYZE

CREATE MATERIALIZED VIEW mw_category_report

AS

select cm.category_id as

CatID,

cm.category_name as CatName,

SUM(od.quantity) as TOTAL_PRODUCT,

COUNT(DISTINCT od.order_id) as TOTAL_ORDERS,

COUNT(DISTINCT od.customer_id) as Unique_Customer,

SUM(od.quantity * od.unit_price) as TOTAL_REVENUE,

EXPERIMENT 8

```
ROUND( SUM(od.quantity * od.unit_price)/COUNT(DISTINCT od.order_id) ,2) as  
AVG_REVENUE,  
(CASE WHEN SUM(od.quantity * od.unit_price) >= 10000 THEN 'Excellent'  
WHEN SUM(od.quantity * od.unit_price) >= 6000 THEN 'Good'  
WHEN SUM(od.quantity * od.unit_price) >= 3000 THEN 'Average'  
ELSE 'Low' END) as Category_performance  
from Order_details as od  
JOIN product_master as pm  
ON od.product_id = pm.product_id  
JOIN category_master as cm  
ON cm.category_id = pm.category_id  
GROUP BY cm.category_id, cm.category_name  
  
SELECT * FROM mw_category_report
```

Learning Outcomes

After completing this experiment, the learner will be able to:

1. Understand the concept and implementation of database Views.
2. Utilize aggregate functions such as COUNT() and DISTINCT to identify and analyze unique customers.
3. Apply joins and filtering techniques to combine information from multiple tables and create meaningful reports.